

Likelihood of Heart Failure based on Health Factors

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I was inclined to choose this dataset for my honors contract due to how heart disease has impacted my family. My family has a history of heart-related illnesses varying from high cholesterol to high blood pressure. I am interested in utilizing ML to determine if there are specific factors that play more of a role in heart failure than others.

The dataset contains 299 samples of patients who have suffered heart failure and went to their follow-up appointments. The 12 features used to determine a person's obesity level are as follows: Age, Anemia, creatinine_phosphokinase, diabetes, ejection fraction, high blood pressure, platelets, serum creatine, serum sodium, sex, smoking, and time between initial visit and follow-up visit (time in days). The label that is assigned for each person is if they passed away during the time period between the initial visit and the follow-up visit (1 = They passed away | 0 = They are alive). For this dataset, I will be using a 75-25 training split meaning that 75 samples will be for testing and 224 samples will be for training. The link for the data repository is here (<https://archive.ics.uci.edu/dataset/519/heart+failure+clinical+records>).

For this model, I plan on using a DecisionTree for this dataset as it allows me to alter the number of splits that are made and view which features were involved in the splits. This will allow me to better understand which features have more of an impact on causing a death by heart failure. I plan on changing two main factors in this project: impurity calculations and number of splits. This way I can determine which loss function works the best for my data and determine if there are several factors that play a role in fatality by heart failure or if there are only a select few. I plan on creating 6 different models to test these: (1) DecisionTree(criterion = "gini", max_depth = 2, randome_state=0), (2) DecisionTree(criterion = "gini", max_depth = 4, randome_state=0), (3) DecisionTree(criterion = "entropy", max_depth = 2, randome_state=0), (4) DecisionTree(criterion = "entropy", max_depth = 4, randome_state=0), (5) DecisionTree(criterion = "log_loss", max_depth = 2, randome_state=0), (6) DecisionTree(criterion = "log_loss", max_depth = 4, randome_state=0). These 6 trials are also conducted on 70-30, 75-25, and 80-20 data splits to see if the change in training splits results in any positive change in the results.

For the 70-30 train-test split trials, the average for each metric are as follows: accuracy has an average of 0.8611111111111111, precision has an average of 0.86060111876999, recall has

an average of 0.86111111111111, and f1 score has an average of 0.85571717145701. For the 75-25 train-test split trials, the average for each metric are as follows: accuracy has an average of 0.83333333333333, precision has an average of 0.83507162974459, recall has an average of 0.83333333333333, and f1 score has an average of 0.82751079462452. For the 80-20 train-test split trials, the average for each metric are as follows: accuracy has an average of 0.80833333333333, precision has an average of 0.81540674587637, recall has an average of 0.80833333333333, and f1 score has an average of 0.80070971826527. All of the charts comparing these scores are located in the 'resultcharts' folder.

When looking at the results of this experiment, there were a couple things that surprised me. The first being the improvement in all accuracy metrics as the training size decreased as I assumed that with the increase in accuracy would be associated with a larger training size since it would lead to more accurate predictions. Maybe in future tests, I can use training splits around the 70-30 range at least for tabular data. Along with that I was interested in how the results differed based on criterion and max depth. When using the gini index, max depth of 4 ended up having lower metrics than with a max depth of 2. The opposite is true with the log loss and entropy criterion as the decision trees yielded higher metric values with a max depth of 4 than with a max depth of 2. I also found it interesting that all the decision trees with a max depth of 2 had the same metric values regardless of the criterion used. It is only when the max depth of the decision tree increases where there is more variance in all metric values.

Lastly, I was surprised at how the decision tree diagrams varied between training splits and trials. I was able to find common factors that determined the likelihood of heart failure: such as time in between visits, ejection fraction, sex, and serum creatinine. However, when the max depth of the decision tree there are a mix of factors such as age, creatinine phosphokinase, and serum sodium. Some future considerations may be to find more data for training to see if that will lead to improved training performance and testing a larger max depth since any max depth larger would create hard-to-read diagrams. Here is the order of the most important features when determining heart failure (from most important to least important): Time (days from initial to follow-up appointment), serum creatinine, ejection fraction, serum sodium, age (in years), sex, creatinine phosphokinase. All other features not mentioned were not used in any of the decision trees.

GitHub with code and diagrams (<https://github.com/JunaydL/CSE475HonorsProject/>).